# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 7 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** $\frac{1}{3}$ of a wall is already painted. Then $\frac{1}{2}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

**(a)** $\frac{1}{6}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{7}{12}$

**2.** Two complementary angles differ by $20°$. The larger angle is: 1

**(a)** $65°$ **(b)** $45°$ **(c)** $75°$ **(d)** $55°$

**3.** For every whole number $n$, $n(n+1)$ is: 1

**(a)** even **(b)** a perfect square **(c)** odd **(d)** a multiple of 3

**4.** Sides $11$ cm, $6$ cm and $5$ cm are given. Which statement is true? 1

**(a)** a triangle can be formed **(b)** only a right triangle is possible

**(c)** no triangle can be formed **(d)** only an equilateral triangle is possible

**5.** If $0 < x < 1$, then $\frac{1}{x}$ is: 1

**(a)** less than 1 **(b)** greater than 1 **(c)** negative **(d)** equal to 1

**6.** Two angles of a linear pair differ by $20°$. The smaller angle is: 1

**(a)** $70°$ **(b)** $90°$ **(c)** $100°$ **(d)** $80°$

**7.** If $1 + 2 + 3 + \cdots + n = 231$, then $n$ is: 1

**(a)** 21 **(b)** 23 **(c)** 20 **(d)** 22

**8.** In an isosceles triangle with perimeter $34$ cm the equal sides are $12$ cm each. The base (cm) is: 1

**(a)** 12 **(b)** 11 **(c)** 9 **(d)** 10

**9.** $\frac{3}{5}$ of $3\frac{1}{3}$ is: 1

**(a)** $4$ **(b)** $2$ **(c)** $1$ **(d)** $3\frac{14}{15}$

**10.** One angle of a linear pair is 2 times the other. The larger angle is: 1

**(a)** $140°$ **(b)** $110°$ **(c)** $130°$ **(d)** $120°$

**11.** The number of reverse-and-add steps for $59$ to become a palindrome is: 1

**(a)** 3 **(b)** 4 **(c)** 2 **(d)** 5

**12.** The two acute angles of a right triangle are in the ratio $1 : 2$. The smaller acute angle is: 1

**(a)** $50°$ **(b)** $20°$ **(c)** $40°$ **(d)** $30°$

**13.** If $x \times \frac{3}{4} = \frac{9}{16}$, then $x \div \frac{3}{4}$ is: 1

**(a)** $1$ **(b)** $1\frac{1}{2}$ **(c)** $2$ **(d)** $\frac{1}{2}$

**14.** If $l \parallel m$, the bisectors of a pair of alternate interior angles are: 1

**(a)** coincident **(b)** parallel **(c)** perpendicular **(d)** intersecting

**15.** Which expression is a multiple of 9 for every digit pair $(a, b)$ with $a > b$? 1

**(a)** $a + b$ **(b)** $(10a + b) + (10b + a)$ **(c)** $(10a + b) - (10b + a)$ **(d)** $10a + b$

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** Show that $n(n + 1)$ is even, and verify for $n = 13$. 2

**17.** In isosceles $\triangle ABC$ with $AB = AC$, side BC is extended to D and $\angle ACD = 115^\circ$. Find $\angle A$ and $\angle B$. 2

**18.** After $\frac{2}{7}$ of a number is removed, 35 is left. Find the number and the part removed. 2

**19.** Two supplementary angles differ by $18^\circ$. Find both angles and the complement of the smaller one. 2

**20.** Using each of the digits $7, 3, 4, 1$ once, make the largest 4-digit number divisible by 3. 2

## SECTION — C (Short Answer Type) 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** A cyclist rides at $\frac{9}{2}$ km per hour for 2 hours, then at $\frac{15}{4}$ km per hour for 4 hours. Find the total distance and the average speed for the whole journey. 3

**22.** Two adjacent angles form a linear pair and one of them is $40^\circ$. Find the other angle, the half of each angle and the angle between the two bisectors. 3

**23.** In how many ways can $15$ be written as the sum of two or more consecutive positive whole numbers? List them. 3

**24.** Two sides of a triangle are $8$ cm and $21$ cm. The third side is a whole number of cm divisible by $5$. List all possible lengths and the largest perimeter. 3

OR

A pole of 20 m is cut into pieces of $1\frac{2}{3}$ m each. How many pieces are obtained and how much is left? 3

**25.** Angles on one side of a straight line at a point are $20^\circ$, $75^\circ$, $30^\circ$ and a fourth angle. Find the fourth angle and name the kind of each angle. 3

OR

The number $7 * 2 * 1$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** A transversal cuts two parallel lines and two interior angles on the same side are in the ratio $2 : 3$. Find these two angles and say how many of the eight angles measure $72^\circ$ and how many $108^\circ$. 5

OR

Find all two-digit numbers that are exactly 2 times the sum of their digits. 5

**27.** $\triangle ABC \cong \triangle DEF$ with $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. Given $\angle A = (3x + 10)^\circ$, $\angle D = (5x - 20)^\circ$, $AB = (2y + 4)$ cm and $DE = (3y - 6)$ cm. Find $x$, $y$, $\angle A$ and $AB$. If $\angle B = 40^\circ$, find $\angle F$. 5

**OR**

Pipe A fills a tank in 6 hours, pipe B in 12 hours, and a drain empties the full tank in 24 hours. Find the fraction filled in 1 hour with all three open, and the time to fill the empty tank. 5

**END OF QUESTION PAPER** **Total: 50 Marks**

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